

Moaz Osama Elkholy

Communication & Electronics Engineer

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Summary

Motivated Communication and Electronics student with a strong foundation in IoT, Networking, Embedded Systems and AI. Experienced in developing and simulating projects using ESP32, STM32, Packet Tracer, C, and Python. Skilled in designing connected systems, troubleshooting networks, and integrating hardware with software. Passionate about learning emerging technologies and applying technical expertise and problem-solving skills in real-world engineering environments.

Skills

Technical Skills: IoT (ESP32, MQTT, HiveMQ, ThingSpeak, Blynk), Networking (Cisco Packet Tracer, DHCP, VLANs, Routing & Switching, Subnetting, HCIA–DataCom concepts), and Embedded Systems (Arduino, STM32, SPI/I2C/UART, FreeRTOS, ADC/GPIO, Sensors, PID Control). Skilled in Programming (Python, C, C++, Java, MATLAB, HTML, CSS, JavaScript) and Git/GitHub, Proteus, Wokwi Simulator, and Microsoft Office.

Soft Skills: Teamwork, Communication, Problem-Solving, Adaptability, Critical Thinking, Presentation, Negotiation

Education

Canadian International College (CIC) – B.Sc. Communication & Electronics Engineering (2022 - 2027)

GPA: 3.53

Experience & Training

Artificial Intelligence Training — Information Technology Institute (ITI), (Aug–Sep 2026)

- Covered Python, Pandas, NumPy, data preprocessing, EDA, Machine Learning, NLP, and Computer Vision.
- Built an AI Customer Feedback Intelligence System using TF-IDF, Logistic Regression, Random Forest, K-Means, and Silhouette Analysis.
- Applied TensorFlow/Keras concepts including CNNs, transfer learning, data augmentation, convolution, ReLU, and pooling.

CCNA Training — National Telecommunication Institute (NTI), (Jan–Apr 2026)

- Comprehensive training in Networking, IP Addressing, Routing (RIP/OSPF), Switching (VLANs/STP).
- Implemented network services (DHCP, NAT, ACLs) with basic security practices.
- Practical labs in network design, configuration, and troubleshooting.

Cloud Services Training - National Telecommunication Institute (NTI), (Nov 2025–Jan 2026)

- Solid understanding of cloud computing concepts (IaaS, PaaS, SaaS).
- Hands-on experience with virtual machines, storage, and networking resources.
- Knowledge of cloud architecture, scalability, high availability, and load balancing.
- Familiar with cloud security basics (IAM, data protection) and service monitoring.

Embedded Systems Intern — Embedded Meetup (ITI Collaboration) (Sep–Oct 2025)

Team Leader, Seat Controller Project

- Designed and implemented an STM32-based ECU using FreeRTOS and multi-protocol (SPI, I2C, UART).
- Integrated EEPROM (HIL) for profile management and calibration.
- Developed a desktop application (BusSniffer) to monitor and analyze serial communication between multiple microcontrollers.

Cyber Security Training — National Telecommunication Institute (NTI), (Aug–Sep 2025)

- Practical experience in Networking, Web Security, and Penetration Testing.
- Applied OWASP Top 10 to identify and fix web vulnerabilities.
- Used tools such as Burp Suite, Wireshark, Nmap, Metasploit, and Kali Linux.
- Gained solid knowledge of Linux commands and networking fundamentals (TCP/IP, packet analysis).

IoT Training – National Telecommunication Institute (NTI), (Jul–Aug 2025)

- Learned IoT protocols (MQTT, HiveMQ, HTTP, CoAP) and ESP32 programming.
- Designed IoT projects (Smart Temperature Control System) using Wokwi & Master of Things platform.
- Enhanced teamwork, communication, and problem-solving skills.

Introduction to Web Technologies – ITI (Aug–Sep 2024)

- Learned front-end development (HTML, CSS, JavaScript).
- Built responsive, user-friendly websites.

Projects

IoT Remote Control & Monitoring System – NTI Training (Aug 2025)

- Designed and implemented an IoT solution combining device control and real-time data monitoring.
- Developed dual operation modes: Manual Mode (remote control via MQTT) and Automatic Mode (temperature-based automation).
- Used ESP32, Relay, LED, Temperature Sensor, and Wokwi simulator for virtual testing.
- Applied MQTT + HiveMQ for reliable cloud communication.

Gesture-Controlled Car – Academic Project (Nov–Dec 2024)

- Programmed Arduino Nano with MPU6050 & Bluetooth (HC-05) to control motion via hand gestures.
- Utilized Arduino Uno in the robot car integrated with a motor driver and another Bluetooth (HC-05) for wireless communication.
- Applied C programming and sensor integration for real-time control.

Self-Balancing System – Academic Project (Mar–May 2025)

- Built a real-time self-balancing system using Arduino, MPU6050, and PID control.
- Applied control theory to maintain system stability.

Network Design Project – Pixels Egypt (Feb 2025)

- Designed a multi-building network using Cisco Packet Tracer, IP subnetting, VLANs, and DHCP.
- Configured routing protocols (RIP, OSPF) and secured Telnet access.

Certifications & Courses

- Raspberry Pi Interfacing & IoT – Embox (2025)
- HCIA–DataCom | Huawei Technologies – (Aug–Oct 2025)
- SPL01 System Programming in Linux – Udemy (2025)
- IoT Course – ITIDA - EME Innovation Labs (2025)
- Introduction to IoT – Cisco Networking Academy (2025)
- Python Programming – Sprints x Microsoft (2025)
- IoT Development – Udemy (2024)

Volunteering & Activities

- Computer Networks – Pixels Egypt (Jan–Feb 2025)
- IEEE CIC Technical Chapter (2024–2025): Participated in AI training, events, and workshops.
- AI Fundamentals Course (IEEE CIC Technical, May 2025).

Languages

Arabic: Native | English: Excellent (Reading/Writing), Very Good (Speaking/Listening)